

The Judge Is Not Its Twin

Post-training makes a model’s writing more predictable but barely moves its taste, as a judge, toward predictable writing

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Abstract

Language models are now routinely graded by other language models. If post-training makes a model’s own writing more predictable, it may also teach the same model, acting as a judge, to reward predictable writing. Original writing would then go unrewarded, and progress on creativity would be invisible to automated evaluation. We test this worry directly. We follow two open model families, OLMo-2 and Zephyr (7B parameters each), through their public training stages and measure every stage twice, once as a writer of short stories and once as a judge of other stories. As writers, the models drift as feared. Each family’s fully trained model finds the trained stages’ stories more familiar than the base model’s, with base, SFT (supervised fine-tuning) and DPO (preference training) in that order in all ten prompts. A model from the other family finds the trained stories less surprising too, by 7.0 to 9.2 percent per token for OLMo-2 and 24 to 25 percent for Zephyr. As judges, they barely move toward predictable writing. Asked which story is better, a question every judge passes a competence check on, no trained judge’s preference for the more predictable story grows by as much as one point of pick probability at its estimate. A post hoc one-sided 95% upper bound on that growth is 2.7 points on an average pair, about the size of OLMo-2’s base judge’s own lean. Asked which is more creative, no trained judge’s estimate prefers the more predictable story more than its base’s does. Training changes judges in other ways. It strengthens a preference for longer stories when the question is creativity, and a preference for one answer slot. It also breaks “more creative” as a question. Asked which is more creative, a story or the same words in scrambled order, trained judges no longer reliably pick the story. Most sit near chance by defaulting to one answer slot, and Zephyr’s SFT judge picks the scrambled text 87% of the time. A follow-up tried to build a cleaner test, two stories from one writer that differ in predictability but not in quality, and could not. The routes that made this writer’s stories less predictable also broke some of them, often enough to fail a quality floor set in advance.

1. Introduction

When a lab wants to know whether a new model writes better stories, it increasingly asks another model. A model acting as a judge reads two answers and says which is better. Its verdicts now rank models on public leaderboards (Zheng et al., 2023). For tasks with a checkable answer, a judge is a cheap stand-in for a human grader. For creative writing there is no checkable answer, so a sharper question appears. Whose taste is the judge applying?

Post-training is the sequence of steps that turns a raw pretrained model into an assistant. It is known to narrow a model’s writing, so that outputs converge on familiar shapes and familiar phrasing (Mohammadi, 2024; Zhang et al., 2025; Karouzos et al., 2026). The judge is usually a post-trained model too, and it can come from the same family as the writer. If the training that narrowed the writer also narrowed the judge, the judge would reward the very predictability the writer drifted toward. A model that learned to write more originally would then score no better, and possibly worse. The judge would be the writer’s twin.

The worry has partial support. Judges give higher scores than human raters do to text with lower perplexity, that is, text the judge finds easier to predict (Wataoka et al., 2024). A recent study finds that a judge prefers the style typical of AI-written text over the unpredictability of human-written stories (Tutone et al., 2026). Judges also favor answers from models related to them (Li et al., 2025). What no study has done is watch one model family’s writing and judging change together, stage by stage, through training. That is the direct test, and public training checkpoints make it possible.

We ran it on two families that release a checkpoint after every training step. OLMo-2 (Team OLMo, 2024) has four: the base model, supervised fine-tuning (SFT, where the model imitates example answers), direct preference optimization (DPO, where it learns from pairs of answers ranked by preference; Rafailov et al., 2023), and a final stage that adds reinforcement learning on tasks with checkable answers. Zephyr (Tunstall et al., 2023) has three: the Mistral-7B base (Jiang et al., 2023), SFT, and DPO. Every stage wrote short stories, and every stage judged pairs of stories written by the other family.

Finding 1: the writer drifts. Every trained stage writes stories that its own family’s trained model finds more familiar than the base model’s stories, and base, SFT and DPO fall in that order in all ten prompts in both families. A model from the other family also finds the trained stories less surprising, by 7.0 to 9.2 percent per token for OLMo-2 and 24 to 25 percent for Zephyr (Section 5).

Finding 2: the judge barely moves toward predictable writing. Asked which story is better, the question every judge passes a competence check on, no trained judge’s preference for the more predictable story grows by as much as one point of pick probability at its estimate. OLMo-2’s estimates edge the predicted way at DPO and final; Zephyr’s go either way depending on which model scores predictability. Ten prompts leave room for more, so we bound it after the fact: the one-sided 95% upper bound on the growth is 2.7 points on an average pair, about the size of OLMo-2’s base judge’s own lean. The predicted drift is not detected, and the bound says how large it could be. Asked which is more creative, no trained judge’s estimate prefers the more predictable story more than its base’s does. The two families do not agree on a direction, and we claim none (Section 6).

Finding 3: training breaks the word “creative”. Asked which story is more creative, every trained judge fails a basic competence check that both base models pass. On scrambled text, some trained judges stop discriminating and default to one answer slot. One, Zephyr’s SFT judge, prefers a story’s words in scrambled order over the story itself 87% of the time. Asked which story is better, every stage passes (Section 7).

Together, the first two findings make the headline. Training changes how these models write far more reliably than it changes how much they, as judges, prefer predictable writing. The third finding points at a different and more practical hazard. The risk in asking a trained judge about creativity is not that it rewards the predictable. The risk is that the question stops tracking whether the text is even a story.

Section 8 reports a follow-up that tried to remove the main design’s central confound, that each judged pair sets a story from one training stage against a story from another, and could not. Its failure is informative in its own right. In the Zephyr writer, the routes that made stories less predictable also broke some of them, often enough that no clean contrast survived a quality floor set in advance.

2. Background

Judges have known habits. Model judges favor one answer position regardless of content (Wang et al., 2023), prefer longer answers, which length-controlled scoring was built to correct (Dubois et al., 2024), and prefer their own writing (Panickssery et al., 2024). Wataoka et al. (2024) reinterpret that last habit as a preference for text the judge finds easy to predict, whoever wrote it. Bias taxonomies now list this as a “familiarity bias” (Zhou et al., 2026). Tutone et al. (2026) find that a judge rating creative writing prefers AI-typical style over human unpredictability. Li et al. (2025) show that judges favor generators related to them by training data or lineage. Each of these studies examines finished models. None follows a judge through the stages of its own training.

Post-training narrows the writer. Aligned models write text that varies less, word by word and in overall shape, than their base models’ text (Mohammadi, 2024). Zhang et al. (2025) trace this narrowing partly to human preference data, which favors familiar text, and name it typicality bias. Karouzou et al. (2026) trace where diversity is lost along several post-training paths and find the supervised fine-tuning step to be a major site. This paper uses the same idea of typicality, measured as predictability under a reader model (Section 3), on both sides of the question.

Predictability is not quality in either direction. Human writing does not sit at the most probable end of a language model’s distribution (Holtzman et al., 2020), so being hard to predict is not by itself a defect. Yet text sampled to be improbable often is defective. Raising the sampling temperature relates only weakly to judged novelty and more strongly to incoherence (Peeperkorn et al., 2024), and the quality of generated text falls at both ends of the likelihood range (Zhang et al., 2020). Creativity is usually defined as novelty together with value (Runco and Jaeger, 2012), so any test of a judge’s taste for the novel has to hold value fixed. Model judges of creative writing also agree poorly with professional writers (Chakrabarty et al., 2024).

3. Measuring how predictable a story is

To ask whether a judge prefers predictable stories, we first need one number for how predictable a story is. We use a reader model’s surprise. A reader is any language model; we feed it the story one token at a time and record how much probability it gave to each token that actually came next. The surprise at a token is the negative log of that probability, measured in nats (natural-log units). A story’s **surprise score** is the average surprise per token. A lower score means a more predictable story. This is the standard mean negative log-likelihood (NLL), and it is our measure of what the literature calls typicality.

The score has a trap, and we fell into it before we fixed it. Scored alone, the first few tokens of any story are expensive, because the reader has no idea yet what the text is about. That cold-start cost is spread over however many tokens follow, so short texts look unpredictable just for being short. A seven-character OLMo-2 completion, “Silence”, scored 6.2 nats per token this way. The fix is to put the writing prompt in front of the story, exactly as the writer saw it, and to average surprise over the story’s tokens only. Every surprise score in this paper uses that frame. We adopted it in a written amendment before re-scoring, together with the two outcomes we expected (the family-level

drift of Section 5 would survive, and OLMo-2’s drop under a stranger reader would appear at about 7 to 10 percent); both expectations were met.

We use the score in two ways to describe the writers, and in a third way inside the judge analysis (Section 4).

The stranger’s view asks whether a story is predictable to anyone. The reader is the base model of the *other* family, which shares no training with the writer. OLMo-2 stories are read by the Mistral-7B base; Zephyr stories are read by the OLMo-2 base.

The family’s view asks whether training taught this family to find this story familiar. For each story we take the score under the family’s fully trained model and subtract the score under the family’s own base model. A negative value means the trained model finds the story easier to predict than its untrained ancestor did. Because it is a difference between two readers on the same text, any fixed offset between those two readers cancels.

4. Models, stories and judges

Writers and stories. Seven checkpoints wrote stories: OLMo-2-1124-7B at base, SFT, DPO and Instruct (the final stage), and Mistral-7B-v0.1, mistral-7b-sft-beta and zephyr-7b-beta (Zephyr’s DPO and final stage). Each wrote 20 stories for each of ten flash-fiction prompts (for example, “Write a very short story about the last phone call ever made”), in a bare completion frame (**Writing prompt:** {prompt}\n\n**Response:**), at temperature 1.0 and top-p 0.95 (the usual settings for how freely a model samples its next word), with at most 220 new tokens. We call each family’s last checkpoint its final stage, so Zephyr’s DPO model is also its final model. That is 1,400 generations. Eleven were empty, all from OLMo-2, and were dropped, leaving 1,389 stories.

Judges. A judge must work at every stage, including base models that cannot follow instructions. So we never ask a judge to write an answer. We show it two stories and read the probability it assigns to the next token, A or B, after this frame:

Here are two responses to the same writing prompt. Prompt: {prompt}

Response A: {story}

Response B: {story}

The more creative response is Response

Each pair is shown in both orders and the two probabilities are averaged, which removes a fixed slot preference from the measurement. The preference itself remains a habit of the judge, and Section 7 measures it. The result, p_x , is the judge’s probability of choosing story x . A second probe replaces “creative” with “better”. In the code that ran, the substitution produced the string “The more better response is Response”. That wording is ungrammatical, and it is what every “better” result in this paper used; Section 7 re-checks it.

A competence check before any judge is read. A judge that cannot do the task still produces probabilities, and a slope computed from a non-judge means nothing. Worse, “bias appears with training” and “competence appears with training” would draw the same picture. So before reading any judge’s taste, we checked that it can tell a story from a ruined copy of itself. For each prompt, six intact stories were each paired with three ruins: the same words in random order (**shuffled**), the first sentence alone (**stub**), and an intact story written for a different prompt (**off-prompt**). That gives 180 pairs per judge. We wrote the rule down before the competence check ran. A judge

passes if it prefers the intact story in at least 80% of shuffled pairs and the lower end of the 95% Wilson interval (a standard confidence interval for a proportion) exceeds 50%. Only the shuffled test gates. Section 6 explains why the main measurement can still read judges that fail this check under one wording.

The main measurement. Each judge stage then compared 200 pairs of intact stories, 20 per prompt, all written by the other family, so in this measurement no judge saw its own family’s writing. The two stories in a pair always come from different training stages of that family (a base story against a DPO story, say), which matters in Section 6. In plain terms, we ask how far a judge’s pick moves toward a story as that story gets more predictable, with length held fixed. For each judge and criterion we fit

$$p_x = a + b \times (\text{difference in surprise score, } x \text{ minus } y) + c \times (\text{difference in log length}) ,$$

where each story’s surprise score comes from its own family’s base model (the judge, recall, is always from the other family). A negative b means the judge prefers the more predictable story. We fit it three ways. The first is the model as written. The second adds the judge’s own “better” probability as a covariate, which asks whether any predictability preference survives once the judge’s own quality call is held fixed. The third uses the creativity-specific contrast, p_{creative} minus p_{better} , as the outcome. The unit of inference is the prompt, so every interval is a 95% cluster bootstrap over the ten prompts, which resamples whole prompts because stories written for one prompt are not independent.

What was written down in advance, and what was not. Two rules were recorded in a dated design document, with their expected outcomes, before the runs that tested them: the prompt-in-front scoring and the competence rule with its 80% bar. A third came mid-stream. We switched the competence check to “better” after the “creative” check had failed and a small diagnostic had shown “better” working on two judges, but before the full check under “better” and before any main-measurement judge ran. The same amendment set two limits for the main measurement in advance: a judge whose picks barely vary (standard deviation of p_x under .07) or flip with display order (mean disagreement above .45) counts as not judging. The same amendment fixed the .5 threshold for treating “better” as entangled with predictability (Limitations), and the 400-character cut had already been set as a sensitivity check. Four analyses were added after seeing data and are labelled where they appear: the family-specific decomposition (Section 5), and the judge-family scoring of predictability, the stage-difference control and the bound on drift (Section 6).

5. The writer drifts

Table 1. How predictable each stage’s stories are. Stranger’s view column: average surprise per token under the other family’s base model, with the base row in nats and later rows as the percent change from base. Family’s view column, in nats per token for every row: score under the family’s trained model minus score under its base model; negative means the trained model finds the story more familiar.

Family	Stage that wrote the story	Stranger’s view	Family’s view
OLMo-2	base	1.915	+0.434
	SFT	−9.2%	−0.121
	DPO	−7.0%	−0.567

Family	Stage that wrote the story	Stranger’s view	Family’s view
Zephyr	final	−7.7%	−0.581
	base	2.105	+0.243
	SFT	−24.9%	−0.139
	DPO (final)	−24.0%	−0.325

Begin with the family’s view, because it asks the question the twin worry turns on. Did training teach this family to find its own kind of writing familiar? It did. OLMo-2’s trained model finds its base model’s stories harder to predict than the base model itself does (+0.434), and finds each later stage’s stories progressively easier (−0.121, −0.567, −0.581). Zephyr follows the same order (+0.243, −0.139, −0.325). The base stage is the most foreign in all ten prompts, and the order base, then SFT, then DPO holds inside every one of the ten prompts in both families. OLMo-2’s final stage is indistinguishable from its DPO stage (−0.581 against −0.567). From base to DPO the change is −1.000 nats per token for OLMo-2 (95% interval −1.06 to −0.95) and −0.569 for Zephyr (−0.59 to −0.55).

One caution applies at the two ends of the order. A base-stage story is scored partly by the model that wrote it, and so is a final-stage story, and a model finds its own samples easy to predict. That pushes the two end stages apart by construction. The middle of the order is the cleaner evidence. OLMo-2’s SFT and DPO stories were written by neither of their scorers and still fall in order (−0.121, then −0.567), and the stranger’s view below involves no self-scoring at all. Zephyr has only one middle stage, because its DPO model is also its final model, so its order leans more on the ends. The family-specific breakdown in the next paragraph inherits the same caution at its ends; its middle stages also fall in order (OLMo-2 SFT −0.264, DPO −0.542).

A second objection is that this is just generic chatbot style, in which case any heavily trained model would find any trained model’s prose familiar. Part of it is. The other family’s trained model, asked the same question about the same stories, agrees story by story (correlation 0.68 on OLMo-2 stories, 0.80 on Zephyr’s). So, in an analysis added after this overlap appeared in the pilot, we subtracted that shared component story by story. What remains is family-specific, and it still orders the stages. It spans 0.692 nats per token for OLMo-2 and 0.261 for Zephyr, with base the most foreign in all ten prompts and the base-to-DPO drop present in all ten (sign test $p = .002$). Much of the familiarity that varies from story to story is shared chatbot style. How much of the change across stages is family-specific differs by family. Measured as the spread between the least and the most familiar stage, for OLMo-2 it is most of the change (0.692 of 1.015, the spread from base to final), and for Zephyr a little under half (0.261 of 0.569).

The stranger’s view asks a blunter question, whether the writing becomes predictable to anyone. It does, and here the drop arrives almost entirely at SFT; later stages hold it rather than extend it. For Zephyr the drop is large. The OLMo-2 base model finds Zephyr’s SFT stories 24.9% less surprising per token than Zephyr’s base stories (2.105 falls to 1.581) and its DPO stories 24.0% less, with the drop in all ten prompts (sign test $p = .002$ at each stage). OLMo-2’s drift is smaller: 9.2%, 7.0% and 7.7% at SFT, DPO and final, read by the Mistral base. It appears in 9, 9 and 8 of the ten prompts, and every cluster-bootstrap interval excludes zero (in nats per token, −0.178 [−0.248, −0.102], −0.134 [−0.191, −0.064] and −0.148 [−0.205, −0.084]).

The two views part company at DPO. The stranger gives back a sliver (OLMo-2 from −9.2% to

−7.0%, Zephyr from −24.9% to −24.0%), while the family’s own trained model keeps finding the writing more familiar (OLMo-2 from −0.121 to −0.567, Zephyr from −0.139 to −0.325).

The generator side of the twin hypothesis therefore holds in both families. Training makes the writing more predictable, and it makes each family’s trained model find its own family’s writing familiar.

6. The judge barely moves toward predictable writing

Table 2. How much each judge’s preference depends on predictability. Slope b from Section 4, per judge stage, 200 pairs each. A negative slope means the judge prefers the more predictable story. Brackets: 95% cluster-bootstrap interval over prompts; the “holding ‘better’ fixed” column is a point estimate only. The first column is the reading this section leads with; the other three are read under “creative”.

Judge	“better” slope [95% CI]	“creative” slope [95% CI]	“creative”, holding the judge’s own “better” fixed	“creative” minus “better”
OLMo-2 base	−.057 [−.12, −.01]	−.096 [−.18, −.02]	−.022	−.039 [−.07, −.01]
OLMo-2 SFT	−.048 [−.11, +.01]	−.058 [−.16, +.03]	+ .001	−.010 [−.05, +.02]
OLMo-2 DPO	−.064 [−.16, +.03]	−.051 [−.19, +.07]	+ .019	+ .013 [−.04, +.06]
OLMo-2 final	−.069 [−.18, +.03]	−.054 [−.21, +.08]	+ .020	+ .016 [−.03, +.06]
Zephyr base	+ .024 [−.01, +.06]	+ .040 [−.01, +.10]	+ .007	+ .016 [−.01, +.04]
Zephyr SFT	+ .023 [−.02, +.08]	+ .052 [−.00, +.11]	+ .021	+ .029 [+ .01, +.05]
Zephyr DPO	+ .076 [−.03, +.20]	+ .120 [+ .03, +.24]	+ .046	+ .044 [−.01, +.11]

The reading we lead with: “better”. Every judge passes the competence check when asked which story is better (Section 7), so these are the slopes a reader can trust most. If the judge were the writer’s twin, they would be negative and would grow more negative from base to final. OLMo-2’s slopes are negative at every stage. They edge the predicted way at DPO and final (−.064 and −.069, against −.057 at base), after a smaller lean at SFT (−.048). Zephyr’s slopes sit near zero at base and SFT (+.024, +.023) and point away from the predictable story at DPO (+.076). Only OLMo-2’s base interval excludes zero.

How much drift the data allow. Ten prompts give wide intervals, so we asked directly how far each trained judge’s slope could have moved toward the predictable story beyond its own base judge’s. This bound was added after seeing Table 2, and we label it post hoc. Every stage of a family judged the same 200 pairs, so we resampled prompts once per draw and refit the base judge and the trained judge on the same draw (5,000 draws), and took the one-sided 95% upper bound on drift toward the predictable story. We ran it twice, once with the story-family reader (each story’s surprise scored by its own family’s base model, as in Table 2) and once with the judge-family reader (the judge’s family base model, which involves no self-scoring). Under “better”, the bounds are .011, .051 and .064 per nat for OLMo-2’s SFT, DPO and final judges with the story-family reader (.013, .048 and .060 with the judge-family reader), and .019 and .016 for Zephyr’s SFT and DPO judges (.029 and .069 with the judge-family reader). To put that in pick terms, the average judged pair differs in surprise score by 0.40 nats per token for the stories OLMo-2’s judges read (0.45 with the judge-family reader) and 0.33 for Zephyr’s (0.30). In pick terms the largest bound is OLMo-2’s final

judge’s: its pick probability could move toward the predictable story by up to 2.7 points with the judge-family reader (2.6 with the story-family reader). For Zephyr the largest is its DPO judge’s, 2.1 points with the judge-family reader (0.5 with the story-family reader). For scale, OLMo-2’s base judge’s whole lean is about 2.6 points with the judge-family reader ($-.059 \times 0.45$) and 2.3 with the story-family reader ($-.057 \times 0.40$), so the data cannot exclude that lean roughly doubling. The bound is for an average pair; for the most different pairs in the study it is larger. It covers movement toward the predictable story only.

The second reading: “creative”. Trained judges fail the competence check under “creative”, so are these slopes worth reading at all? On the real story pairs they carry information. The scrambled-text failure shows up on word salad, and real pairs contain none. On these 200 pairs every judge’s picks vary (standard deviation of p_x from .14 to .34 under “creative”) and disagree across display orders by .12 to .38 on average, inside both limits set in advance. Two real pairs show what that means for Zephyr’s SFT judge, the one that preferred word salad. For the museum prompt, it compared a base-model story that repeats its own prompt and stops mid-sentence (“I wish”) with a final-stage story about a silver pitcher of similar length (878 and 817 characters). It picked the pitcher story as more creative with probability .87, in both display orders. For the red-light prompt, it preferred a story whose driver runs the light, and which ends in a leaked “Word count: 153”, over one that stays at the light (.88; 564 and 649 characters). We chose these two to illustrate, one sensible pick and one not. On real text these judges produce varied picks within the limits set in advance; whether they pick well is a separate question, and that is why “creative” is the second reading, not the first.

OLMo-2’s judges’ estimates lean toward the more predictable story when asked which is more creative. The lean is largest in the untrained base model ($-.096$, the only OLMo-2 interval that excludes zero), and it is smaller after training ($-.058$, $-.051$, $-.054$). Training does not add to it. Zephyr’s estimates point the other way, toward the less predictable story, and grow with training, from $+.040$ at base to $+.052$ after SFT and $+.120$ after DPO; only the DPO interval excludes zero. The Zephyr pattern survives dropping every pair that contains a story under 400 characters ($+.051$, $+.071$, $+.112$). Under “creative” the drift bounds are smaller still: with the story-family reader, .016 and .026 for OLMo-2’s DPO and final judges and below zero for the other three, and with the judge-family reader at most .035 for any trained judge.

Holding the judge’s own “better” verdict fixed shrinks every “creative” slope to within .05 of zero. The creativity-specific contrast is small everywhere, from $-.039$ to $+.044$, and it does not move toward predictability with training in either family. Two of its intervals exclude zero, and they point in opposite directions. OLMo-2’s base judge leans toward the predictable story and Zephyr’s SFT judge away from it.

We also re-ran every fit on long stories only (at least 400 characters, a sensitivity cut set in advance) and, after seeing the first results, with surprise scored by the judge’s own family base model instead of the story’s (for the stories a judge reads, that is the stranger’s-view reader of Section 3). Under “creative”, no form shows a trained judge preferring the predictable story more than its base does. Under “better”, some trained judges drift the predicted way in some forms. The largest such drift, .016, is Zephyr’s DPO judge with surprise scored by the judge’s own family ($-.016$ against $-.000$ at base), well inside its interval. The predicted pattern, a clear preference for predictable writing that appears or strengthens with training, is not detected.

A caution on sign. In this design every pair sets one training stage’s story against another’s. Because training makes stories more predictable (Section 5), the predictability difference within

a pair is nearly the same thing as the stage difference. After seeing Table 2, we added the stage difference as a covariate. This analysis was not planned, and we label it as post hoc. It was run only on the long-story subset (both stories at least 400 characters), not on all pairs; for OLMo-2’s judges that subset is every pair, and for Zephyr’s it is 152 of 200. There, under “creative”, it moves OLMo-2’s slopes from $-.028$ at base to $+.026$, $+.072$ and $+.079$ for SFT, DPO and final, all intervals including zero. Under “better” it moves them from $-.032$ at base to $-.011$, $+.009$ and $+.011$. Zephyr’s “creative” slopes stay positive under the same control ($+.044$, $+.061$, $+.091$). When one nearly redundant covariate flips a sign, the data cannot separate “rewards less predictable writing” from “rewards later-stage writing”. This paper therefore claims neither sign for the judge slope. It claims only that the predicted drift is not detected, and bounds how large it could be. Separating the two readings needs pairs from one writer at one stage that differ in predictability alone, which is what Section 8 attempted.

7. What training does to a judge instead

The judges are not inert. On the same pairs, the spread of their picks grows from base to final (standard deviation of p_x under “creative” from $.14$ to $.25$ for OLMo-2 and from $.16$ to $.34$ for Zephyr), and several of their habits grow.

The word “creative” stops working. The competence check told us this first (Table 3). Asked for the more creative story, both base models pass, preferring the intact story over its shuffled copy 95% (OLMo-2) and 88% (Zephyr) of the time. Every trained stage fails, in one of two ways. OLMo-2’s three trained stages and Zephyr’s DPO stage mostly answer “A” in both display orders, so they sit near chance. They default to a slot rather than judge. Zephyr’s SFT stage does something stranger. It picks the scrambled text 87% of the time, with 99% of its next-token probability on the two answer letters. It is answering, confidently, and its answer is word salad.

Table 3. Competence check, share of shuffled pairs in which the judge prefers the intact story (60 pairs per judge). Pass requires at least $.80$ and a Wilson lower bound above $.50$.

Judge	asked “more creative”	asked “better”
OLMo-2 base	.95	1.00
OLMo-2 SFT	.47	.97
OLMo-2 DPO	.47	.98
OLMo-2 final	.48	.98
Zephyr base	.88	.97
Zephyr SFT	.13	.88
Zephyr DPO	.40	.98

Asked for the better story, every stage passes. To check that the word, not the formatting, was responsible, we crossed the criterion with the prompt format on the two SFT judges. Wrapping the question in each model’s chat template changed nothing. Changing the word changed everything. OLMo-2 SFT went from $.47$ (bare frame) and $.40$ (chat template) under “creative” to $.97$ and 1.00 under “better”, and Zephyr SFT went from $.13$ and $.07$ to $.88$ and $.88$. Because the “better” probe as run was ungrammatical, we repeated the competence check, after seeing the results, on both SFT judges with the grammatical wording. Both pass. OLMo-2 SFT prefers the intact story in $.95$ of shuffled pairs and Zephyr SFT in $.97$, against $.97$ and $.88$ with the original wording. The typo is not what rescues the judges. Our reading of Zephyr’s SFT judge is that it takes “creative” to mean

“unusual”, with no floor on value, and scrambled text is maximally unusual. The other four trained judges mostly fall back on a slot, though Zephyr’s DPO judge still picks the scrambled text 60% of the time.

This failure runs in the opposite direction from the twin hypothesis. A twin judge would punish the unusual. Zephyr’s SFT judge, asked about creativity, rewards the unusual without regard to whether the text is a story at all, and on scrambled pairs the other trained judges stop discriminating. We report it as a finding of its own (Finding 3) because it is the more immediate hazard for anyone using a small trained model to rate creative writing. The question “which is more creative?” can produce confident answers that no longer track anything a reader would call creative.

Two other checks were reported for every stage. Stub pairs were easy, and every judge preferred the full story over its first sentence at least 95% of the time. Off-prompt pairs were not. Accuracy stayed between .50 and .70 under “better”, so no judge reliably notices that a story answers a different prompt. Our claims are therefore about the text of a story, not about how well it fits its prompt.

Length and position preferences grow. Under “creative”, the slope on length difference rises with training, from +.005 to +.078, +.102 and +.108 for OLMo-2, and from +.121 to +.131 and +.202 for Zephyr, so longer stories win more often. Under “better” the OLMo-2 length slope moves the other way, from $-.069$ at base to $-.158$ at the final stage, while Zephyr’s still favors longer stories and grows (+.071, +.090, +.175). The judges’ preference for one display slot also rises under “creative” from base to final, from .15 to .37 for OLMo-2 and from .15 to .21 for Zephyr (mean disagreement between the two display orders), though not at every step (Zephyr’s SFT judge dips to .12, and OLMo-2’s peaks at DPO, .38). These are the habits a practitioner should correct for. These data give no reason to add a typicality correction on top of them.

8. Why a cleaner test could not be built

Section 6 left one door open. Because every pair in the main design mixes two training stages, a preference for less predictable writing and a preference for later-stage writing cannot be told apart. The clean experiment is easy to state. Take one writer at one stage. Produce pairs of stories for the same prompt, of similar length and similar quality, that differ in predictability. Ask judges from every stage which they prefer. If later judges increasingly prefer the more predictable story, the twin reading comes back.

We ran the first half of that experiment, building the pairs, with Zephyr’s final model (zephyr-7b-beta) as the writer and twelve flash-fiction prompts. This time stories were allowed to finish. The token limit was raised to 768, and all 49 ordinary stories (four per prompt, five for one) ended on their own, the longest at 681 tokens; the asking route was rerun at 1,024 so that its longer stories could finish too. Rules set at the start of this work, and applied mechanically before any story was read or scored, kept only finished stories (a natural stop, final punctuation, at least 100 words, no leaked template text) and paired stories only when the longer was at most 1.25 times the length of the shorter. One rule, the pattern for unfilled placeholders, was narrowed after the first stories were seen; it changed no story’s fate.

Two gates stood between the pairs and any judge. The first is a **manipulation check**, which requires the treated side of each pair to be measurably less predictable to an outside reader (OLMo-2’s base model, prompt in front). The second is a **blind read**, in which the two authors rated every story in four sample pairs together, agreeing on one score per story from 1 to 5 for being a coherent, complete story that fits its prompt, and recorded jointly which story in each pair was better and

which was more creative. The key to which story was which was kept in a separate file that neither rater saw, and the agreed ratings were committed to version control before it was opened. The pass rule was written before the ratings were recorded or unblinded. Every treated story must score at least 3, and the treated stories’ median must be at least 4.

We tried three ways of producing the treated side (Table 4). Asking did not move predictability. Heating and selecting did, but each produced at least one broken story, and each failed the blind read’s floor.

Table 4. Three routes to a less predictable story from the same writer. Treated side: the story the route was meant to make less predictable (for selecting, the less predictable story of a pair chosen by natural spread). In one heated pair it came out more predictable (see below the table). Gap: average surprise per token, treated side minus partner, under the OLMo-2 base reader, with a 95% bootstrap interval. Blind read: coherence scores (1 to 5) for the four sampled pairs.

Route	How the treated story was made	Gap in surprise (nats per token)	Blind read, treated side	Blind read, partner
Asking	an added instruction to write a response with under 10 percent probability	−0.031 [−0.097, +0.037]; 11 of 29 pairs in the intended direction	not run	not run
Heating	sampling temperature raised from 1.0 to 1.3	+0.768 [+0.595, +0.908]; 19 of 20 pairs	2, 3, 4, 4	5, 5, 5, 5
Selecting	ordinary stories at temperature 1.0, paired by natural spread	+0.176 [+0.053, +0.302], measured on held-out tokens	4.5, 1, 2, 4	2, 3.5, 4, 3

In one of the four heated pairs read blind, the heated story (scored 3) was in fact the more predictable of the two (1.355 against 1.710 nats per token).

Asking did not move predictability. Told to write a story it would give under 10 percent probability, the writer produced stories that were, if anything, slightly more predictable, and longer. The reader can see real differences. With the same frame it had separated Zephyr’s training stages in the main study (Section 5), and it registers the heating route’s shift below. Our reading is that, asked for the improbable, the writer reached for melodrama (apocalypses, last calls, hidden rooms), which may be exactly what a base model expects.

Heating moved predictability and broke a story. At temperature 1.3 the gap was large and consistent. The blind read failed both legs of its rule, because one heated story scored 2 and the median was 3.5. That story contradicts itself. A river that “hasn’t existed for centuries” is also, in the same story, “missing from your map”. The raters preferred the ordinary story as better and as more creative in all four pairs.

Selecting gave a real but smaller spread, and its widest pairs failed the same floor. Ordinary stories for one prompt already vary in predictability. Measuring that spread fairly takes care, because if pairs are chosen for having the widest gap, the choosing inflates the gap. So we chose pairs using one half of each story’s tokens and measured the gap on the other half. The held-out gap was +0.176 nats per token, with 7 of 11 prompts positive (one prompt had no pair

within the length rule). About a third of the naive gap was selection, since the same pairs measured on the tokens used to choose them gave 0.260. The interval overstates our confidence. On synthetic banks of pure noise, the same procedure declared a gap real 9 times in 150 (6%, where the lower end of a two-sided 95% interval should clear zero by chance about 2.5% of the time). Roughly, since the quantities are measured differently, the natural spread is a quarter of the temperature gap and a third of the drift from Zephyr’s base to its SFT stage. For the blind read we deliberately took the widest pair from each of four prompts as a stress test, since if quality falls with predictability, it should show most there. Two of the four less predictable stories scored 1 and 2. Both contain logic breaks. In one, an inherited debt turns into a thriving business, and in the other a clock is a neon readout in one sentence and a metal ball in the next. The other two scored 4 and 4.5, and the raters picked the 4.5 as both better and more creative. The one low score on the predictable side (a 2) reads fluently and plausibly misses its prompt. None of the eight stories repeats itself (the share of three-word sequences that repeat an earlier one is at most 0.032 in each), so this is not the high-likelihood repetition failure that Zhang et al. (2020) describe.

Both reads together. Counting across both reads by each pair’s measured predictability rather than by route, three of the eight less predictable stories scored 2 or below, all with logic breaks, against one of the eight more predictable stories. Median coherence was about the same on the two sides (4 on the less predictable side, 3.75 on the more predictable). The raters picked the more predictable story as better in 6 of 8 pairs and as more creative in 6 of 8. These are agreed scores from one pair of raters on eight pairs, four of them chosen to be extreme, so we report counts, not a test. What the counts do show is that neither route delivered a set of less predictable stories that reliably held together, which is what the floor demanded.

Appendix A checks whether the reader’s surprise falls on the broken sentences themselves. In the three broken stories it does not concentrate there.

What this means for the judge question. The question Section 6 left open, whether a judge prefers the more predictable of two stories when quality is held fixed, could not be asked with the pairs we could build, because in them predictability and quality did not come apart cleanly. Some less predictable stories broke, and the two human raters mostly preferred the more predictable story. A judge that did the same on these pairs might be making the same defensible call. No judge was run on any of these pairs. Answering the question needs less predictable stories that reliably hold together, from a stronger writer or from human-written stories with human quality ratings. Temperature is a poor lever for this, in line with Peeperkorn et al. (2024).

9. Discussion

The twin worry is reasonable, and on the writer’s side it is borne out. Post-training narrows these writers, and each family comes to find its own narrowed writing familiar. The same training barely moves these models, as judges, toward predictable writing, and a one-sided bound says how far it could have. What training does teach them is a set of habits already known from larger judges: favoring longer stories (under “creative” in both families, and under “better” in Zephyr), favoring a slot, and, most sharply, losing any stable meaning for “more creative”. So for these two families, the practical risk in using a small trained model to judge creative writing is not a hidden taste for predictability. It is the ordinary, measurable habits, plus a criterion word that no longer means what the person asking intends.

The follow-up adds a second lesson, about measurement. A surprise score is often read as a proxy for novelty. In this writer, both routes that lowered predictability also produced broken stories,

often enough to fail a quality floor set in advance. A study that manipulates predictability and then asks whether judges reward it has to show, independently, that its less predictable items are still good. Otherwise a judge that “penalizes the unpredictable” may simply be penalizing the incoherent, and it would be right to.

Limitations

- **Scale.** Two families at 7B parameters, one size each. The planned third family (Tülu-3, built on Llama-3.1) was not run.
- **Prompts and sample.** Ten flash-fiction prompts in the main study, twelve in the follow-up, 200 judged pairs per judge. The prompt is the unit of inference, so intervals rest on ten clusters, and a percentile bootstrap over so few clusters can understate uncertainty; the drift bound may be optimistic for that reason.
- **Unfinished stories in the main study.** Stories were cut at 220 new tokens and the stopping reason was not recorded. Some main-study stories end mid-sentence, and judges saw them as written. The cut falls on the longest stories, so it is tangled with the judges’ preference for length (Section 7); we did not separate the two.
- **The “better” probe’s wording.** Every “better” result used “The more better response is Response”. Every stage passed the competence check with it. The grammatical version, run on the two SFT judges, also passes (.95 and .97 on shuffled pairs); the other five stages and the main measurement were not re-run with it.
- **Prompt fit.** No judge reliably detects an off-prompt story (.50 to .70 under “better”), so our claims concern the text, not its fit to the prompt.
- **Self-scoring at the ends of the family’s view.** Base-stage and final-stage stories are scored partly by their own writer, which widens the gap between the end stages by construction (Section 5). The judge analysis inherits a milder form, since its surprise score for a base-stage story comes from the model that wrote it. Scoring surprise with the judge’s own family instead, which involves no self-scoring, leaves the “creative” conclusion unchanged (no trained judge prefers the predictable story more than its base does), though Zephyr’s DPO slope shrinks from +.120 to +.029. Under “better” it moves the drift bounds both ways: OLMo-2’s shrink slightly, and Zephyr’s DPO bound grows from 0.5 to 2.1 points (Section 6).
- **Sign of the judge slope.** Pairs cross training stages, so the direction of any judge preference is confounded with stage (Section 6). We claim only that the predicted drift is not detected, with a post hoc bound on its size.
- **No judge read its own family.** Every judge read the other family’s stories, so this study tests whether training gives a judge a taste for predictable writing. It does not test whether a judge favors its own family’s stories, the self-preference that Panickssery et al. (2024) and Li et al. (2025) study.
- **No human judgment of creativity in the main study.** “Better” is the judge’s own quality axis. It could be polish rather than value. Its correlation with the predictability gap stayed between .11 and .29 in absolute value at every stage, under the .5 threshold we set in advance for treating it as entangled.
- **The follow-up.** One writer, two raters (the authors) giving one agreed score per story, so no inter-rater agreement can be reported, eight pairs (four natural pairs, the widest by design, and four sampled heated pairs), and half-point scores. Nothing statistical is claimed from it, and the localization check covers three stories.

10. Conclusion

We asked whether a model’s judgment drifts with its writing as training proceeds. For OLMo-2 and Zephyr, the writing drifts. Every trained stage writes more predictably than its base, and each family comes to find its own writing familiar. The judgment barely moves toward predictable writing: asked which story is better, no trained judge’s preference for predictable stories grows by as much as a point of pick probability at its estimate, and a post hoc one-sided 95% upper bound on that growth is 2.7 points on an average pair. Training instead makes judges more swayed by length and position, and it breaks “creative” as a question, so that trained judges asked for creativity no longer reliably prefer a story to its own scrambled words. The cleaner test that could settle the sign of a judge’s taste needs less predictable stories that reliably hold together. Pushing the Zephyr writer toward them broke too many to pass a quality floor, which is a warning for anyone who reads low predictability as novelty.

Appendix A. Where the reader’s surprise sits

If a high surprise score meant a broken story, the broken sentences themselves should be the most surprising ones. We checked that in the three broken stories. Before any sentence was scored, the break sentences in the three logic-broken stories were marked from the texts alone, seven sentences in all, and the marks were committed. Then each sentence was scored for average surprise. One of the seven break sentences fell in its story’s most surprising quarter, where chance would put about two, and four of the seven sat above their story’s median sentence. By the reading rule written with the marks, that pattern is diffuse, so the breaks are not where the reader’s surprise concentrates. Among the most surprising sentences were openings and odd premises: “a bill for over a thousand cosmic surgeries”, “a deep blue river with no name”, a clock that “blinked 12:00 in neon red”. Three stories cannot carry a claim, and an opening sentence has the least context, so part of its surprise is position. One hypothesis for later work is that a reader’s surprise may respond more to an odd premise than to a contradiction, so that selecting for low predictability selects stories that start somewhere odd, whether or not they carry it through.

Author contributions

A.N.K. designed the study, ran all generation, scoring, judging and analysis, and co-rated the stories in both blind reads. A.B. (M.S. in Electrical and Computer Engineering, University of Rochester) co-rated the stories in both blind reads.

AI use

The text of this paper was drafted with Claude (Anthropic), an AI assistant, and reviewed and revised by the authors.

Code and data

Code, generated stories, predictability scores, judge outputs and the dated design documents are at <https://github.com/IamArmanNikkhah/judge-is-not-its-twin>.

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